

Sentiment Analysis System Project Report

Introduction

The Sentiment Analysis System is a natural language processing (NLP) application designed to analyze sentiment in textual data. It categorizes sentiments into three categories: positive, negative, and neutral. The system is capable of visualizing the sentiment results based on factors such as gender, age, language, and city. It integrates seamlessly with Google Sheets for data collection via Google Forms and operates as a web application accessible over a LAN.

Objectives

1. Analyze textual data to predict sentiment categories (positive, negative, neutral).
2. Provide insights through visualizations based on demographic and linguistic factors.
3. Simplify data collection and integration by utilizing Google Sheets.
4. Offer a user-friendly web interface for accessing and visualizing results.

Use Cases

- **Product/Service Monitoring** - Understand customer sentiment towards products or services.
- **Survey Analysis** - Analyze sentiment from survey responses.
- **Social Media Monitoring** - Evaluate public sentiment on social media platforms.
- **Feedback Analysis** - Process and analyze feedback for actionable insights.

Implementation

Backend

1. **Libraries and Tools:**
 - `google_auth_oauthlib` and `googleapiclient` for Google Sheets API integration.
 - `vaderSentiment` for sentiment analysis.
 - `pandas` for data manipulation.
2. **Sentiment Analysis:**
 - Text from a specified column is analyzed using VADER Sentiment Analysis.
 - Sentiment is categorized based on compound scores: Positive, Neutral, or Negative.

3. **Google Sheets Integration:**
 - Authenticates using OAuth 2.0.
 - Retrieves data dynamically for real-time analysis.
4. **Data Storage:**
 - Results are saved as a CSV file (`results.csv`) for reuse and visualization

Frontend

1. **Streamlit Web Application:**
 - Provides a sidebar menu with three options: Home, Analysis, and Results.
 - User-friendly inputs for Google Sheet ID, data range, and column name.
2. **Dynamic Visualizations:**
 - Built with `plotly`, `matplotlib`, and `seaborn`.
 - Options include:
 - i. Pie charts for sentiment distribution.
 - ii. Histograms for column data.
 - iii. Bar plots for categorical data distribution.
 - iv. Count plots for sentiment distribution by category.
 - v. Scatter plots for sentiment score comparisons.

Detailed Implementation

Workflow

1. User selects "Analysis" and provides the Google Sheet ID, data range, and column for analysis.
2. The system retrieves the data and performs sentiment analysis using VADER.
3. Results are saved to a CSV file.
4. In the "Results" section, users can view the data and generate visualizations.

Features

1. **Home Page:**
 - Displays an overview of the application.
 - Explains sentiment analysis and its visualizations.
2. **Analysis Section:**
 - Connects to a Google Sheet via its ID.
 - Performs sentiment analysis on a specified column.
 - Saves results to a CSV file for further use.
3. **Results Section:**
 - Displays the analyzed data.
 - Provides multiple visualization options, such as pie charts, histograms, bar plots, count plots, and scatter plots.

Backend code for sentiment detection:

```
import streamlit as st

from google_auth_oauthlib.flow import InstalledAppFlow

from googleapiclient.discovery import build

from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer

import pandas as pd

import plotly.express as px

import matplotlib.pyplot as plt

import seaborn as sns

st.set_page_config(page_title="Sentiment Analysis System")

st.title("SENTIMENT ANALYSIS SYSTEM")

choice=st.sidebar.selectbox("My Menu", ("HOME", "ANALYSIS", "RESULTS"))

if(choice=="HOME"):

    st.image("https://user-images.githubusercontent.com/57702598/90991088-264c8880-e56c-11ea-9895-90029d3c2139.gif")

    st.write("It is a natural language processing system which can analyze the sentiment of text data")

    st.write("The application predicts the sentiment into 3 categories: Positive, Negative and Neutral")

    st.write("This application then visualises the results based on different factors such as age, gender, language and city")

elif(choice=="ANALYSIS"):

    sid=st.text_input("Enter your Google Sheet ID")

    r=st.text_input("Enter range between first column and last column")
```

```

c=st.text_input("Enter column name that is to be analyzed")

btn=st.button("Analyze")

if btn:

    if 'cred' not in st.session_state:

        f=InstalledAppFlow.from_client_secrets_file("key.json",
["https://www.googleapis.com/auth/spreadsheets"])

        st.session_state['cred']=f.run_local_server(port=0)

    mymodel=SentimentIntensityAnalyzer()

    service=build("Sheets", "v4",
credentials=st.session_state['cred']).spreadsheets().values()

    d=service.get(spreadsheetId=sid, range=r).execute()

    data=d["values"]

    df=pd.DataFrame(data=data[1:], columns=data[0])

    l=[]

    for i in range(0, len(df)):

        t=df._get_value(i, "Review")

        pred=mymodel.polarity_scores(t)

        if(pred['compound']>0.5):

            l.append("Positive")

        elif(pred["compound"]<-0.5):

            l.append("Negative")

        else:

            l.append("Neutral")

    df['Sentiment']=l

    df.to_csv("results.csv", index=False)

```

```
st.subheader("The analysis results have been saved by the name of a
results.csv")

elif(choice=="RESULTS"):

    df=pd.read_csv("results.csv")

    choice2=st.selectbox("Choose visualization", ("NONE", "PIE CHART", "HISTOGRAM",
"BAR PLOT", "COUNT PLOT", "SCATTER PLOT"))

    st.dataframe(df)

    if(choice2=="PIE CHART"):

        sentiment_counts = df['Sentiment'].value_counts()

        plt.figure(figsize=(6, 6))

        sentiment_counts.plot.pie(colors=['green', 'orange', 'red'])

        plt.title('Sentiment Distribution')

        plt.ylabel('')

        st.pyplot(plt)

    if(choice2=="HISTOGRAM"):

        k=st.selectbox("Choose column", df.columns)

        if k:

            plt.figure(figsize=(6, 4))

            sns.histplot(df[k], bins=5, kde=True, color='blue')

            plt.title(k+' Distribution')

            plt.xlabel(k)

            plt.ylabel('Frequency')

            st.pyplot(plt)

    if(choice2=="BAR PLOT"):

        g=st.selectbox("Choose column", df.columns)
```

```

if g:

    gender_counts = df[g].value_counts()

    plt.figure(figsize=(6, 4))

    sns.barplot(x=gender_counts.index, y=gender_counts.values)

    plt.title(g+' Distribution')

    plt.xlabel(g)

    plt.ylabel('Count')

    st.pyplot(plt)

if(choice2=="COUNT PLOT"):

    h=st.selectbox("Choose column", df.columns)

    if h:

        plt.figure(figsize=(8, 6))

        sns.countplot(data=df, x='Sentiment', hue=h, palette='Set2')

        plt.title('Sentiment Distribution by '+h)

        plt.xlabel('Sentiment')

        plt.ylabel('Count')

        plt.legend(title=h)

        st.pyplot(plt)

if(choice2=="SCATTER PLOT"):

    i=st.selectbox("Choose column", df.columns)

    j=st.selectbox("Choose another column", df.columns)

    if i and j:

        plt.figure(figsize=(8, 6))

```

```
df['SentimentScore'] = df['Sentiment'].map({'Positive': 1, 'Neutral': 0,
'Negative': -1})

plt.figure(figsize=(8, 6))

sns.scatterplot(data=df, x=i, y='SentimentScore', hue=j, style=j, s=100)

plt.title('Scatterplot: '+i+' vs Sentiment Score')

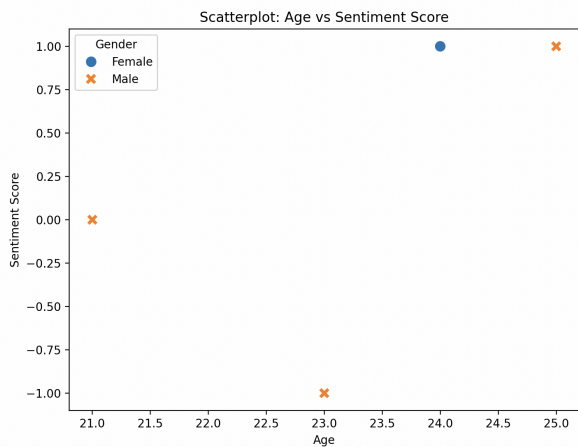
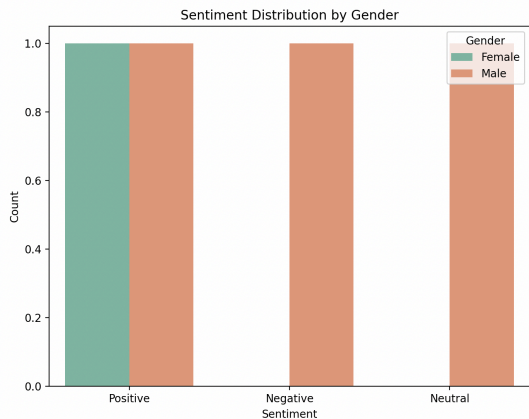
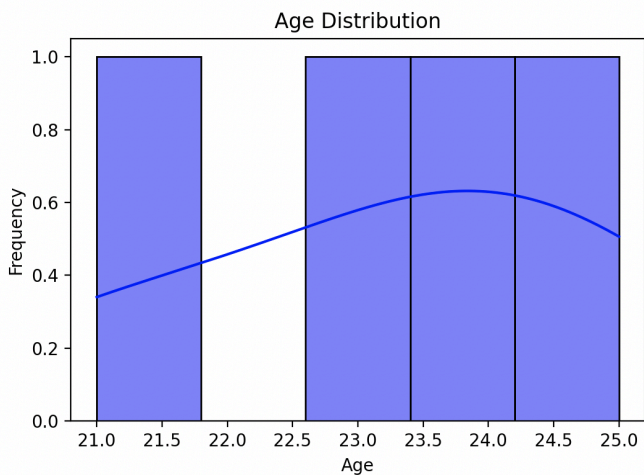
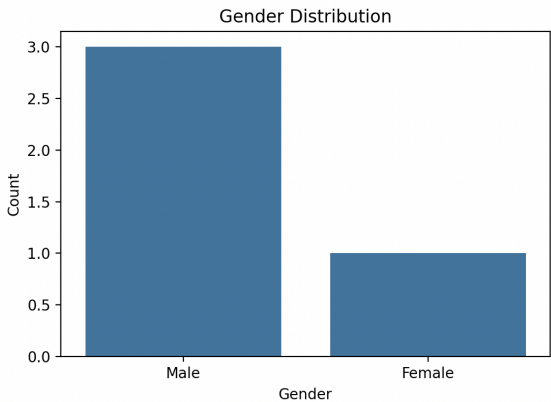
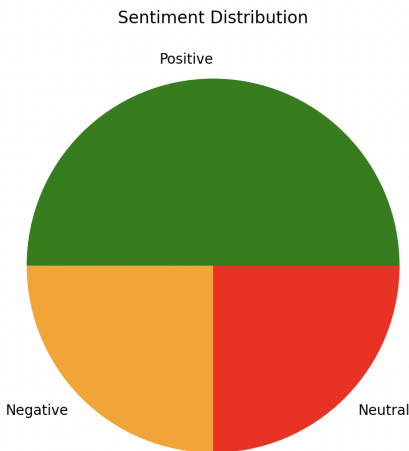
plt.xlabel(i)

plt.ylabel('Sentiment Score')

plt.legend(title=j)

st.pyplot(plt)
```

Data analysis charts:



Results and Analysis:

My Menu

HOME

SENTIMENT ANALYSIS SYSTEM

The filet mignon was remarkable.

It is a natural language processing system which can analyze the sentiment of text data

The application predicts the sentiment into 3 categories: Positive, Negative and Neutral

This application then visualises the results based on different factors such as age, gender, language and city

My Menu

ANALYSIS

SENTIMENT ANALYSIS SYSTEM

Enter your Google Sheet ID

1wAF08_f7lj_ePgO2iUTSyHHuBy37r637HDkGpbr1JDg

Enter range between first column and last column

B:H

Enter column name that is to be analyzed

Review

Analyze

The analysis results have been saved by the name of a results.csv

My Menu

ANALYSIS



SENTIMENT ANALYSIS SYSTEM

Enter your Google Sheet ID

1wAF08_f7Lj_ePgO2lUtSyHHuBy37r837HDkGpbr1JDg

Enter range between first column and last column

B:H

Enter column name that is to be analyzed

Review

Analyze

The analysis results have been saved by the name of a results.csv

My Menu

RESULTS



SENTIMENT ANALYSIS SYSTEM

Choose visualization

NONE



	Timestamp	Name	Age	Gender	Language	Place	Review
0	1/20/2025 0:38:10	sumana	24	Female	English	Hyderabad	The product was amazing
1	1/20/2025 0:38:31	Nadeesh	25	Male	English	Vijayawada	The product was too good, loved
2	1/20/2025 0:39:15	Vipul	23	Male	English	Hyderabad	The quality of the product sucks.
3	1/20/2025 0:39:51	Kalyan	21	Male	English	Guntur	There is some problem with batt

My Menu

RESULTS

My Menu

RESULTS

SENTIMENT ANALYSIS SYSTEM

Choose visualization

NONE

NONE

PIE CHART

HISTOGRAM

BAR PLOT

COUNT PLOT

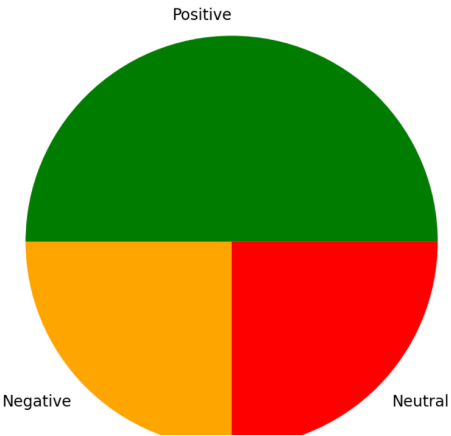
SCATTER PLOT

Choose visualization

PIE CHART

	Timestamp	Name	Age	Gender	Language	Place	Review
0	1/20/2025 0:38:10	sumana	24	Female	English	Hyderabad	The product was amazing
1	1/20/2025 0:38:31	Nadeesh	25	Male	English	Vijayawada	The product was too good, loved
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3	1/20/2025 0:39:51	Kalyan	21	Male	English	Guntur	There is some problem with batt

Sentiment Distribution



My Menu

RESULTS

My Menu

RESULTS

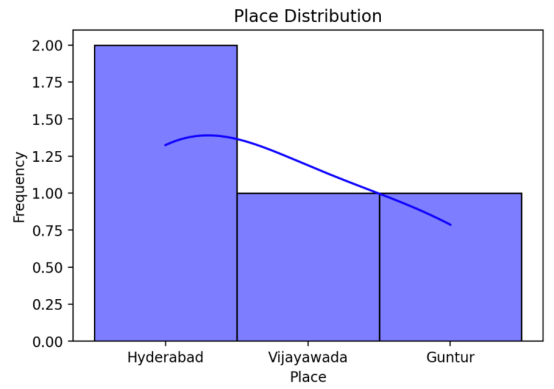
Choose visualization

HISTOGRAM

	Timestamp	Name	Age	Gender	Language	Place	Review
0	1/20/2025 0:38:10	sumana	24	Female	English	Hyderabad	The product was amazing
1	1/20/2025 0:38:31	Nadeesh	25	Male	English	Vijayawada	The product was too good, loved
2	1/20/2025 0:39:15	Vipul	23	Male	English	Hyderabad	The quality of the product sucks.
3	1/20/2025 0:39:51	Kalyan	21	Male	English	Guntur	There is some problem with batt

Choose column

Place



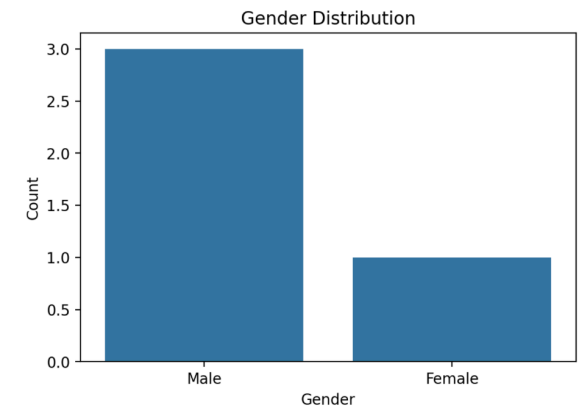
Choose visualization

BAR PLOT

	Timestamp	Name	Age	Gender	Language	Place	Review
0	1/20/2025 0:38:10	sumana	24	Female	English	Hyderabad	The product was amazing
1	1/20/2025 0:38:31	Nadeesh	25	Male	English	Vijayawada	The product was too good, loved
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Choose column

Gender



My Menu

RESULTS

Choose visualization

BAR PLOT

	Timestamp	Name	Age	Gender	Language	Place	Review
0	1/20/2025 0:38:10	sumana	24	Female	English	Hyderabad	The product was amazing
1	1/20/2025 0:38:31	Nadeesh	25	Male	English	Vijayawada	The product was too good, loved
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3	1/20/2025 0:39:51	Kalyan	21	Male	English	Guntur	There is some problem with batt

Choose column

Gender

Timestamp

Name

Age

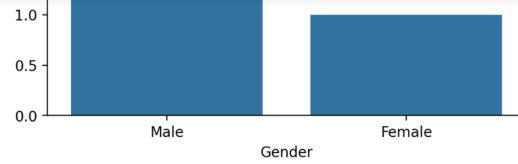
Gender

Language

Place

Review

Sentiment



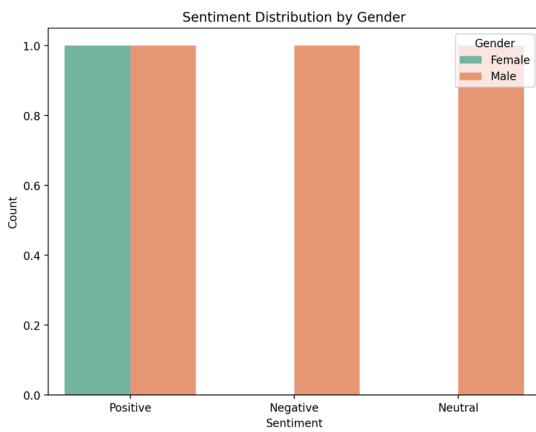
Choose visualization

COUNT PLOT

	Timestamp	Name	Age	Gender	Language	Place	Review
0	1/20/2025 0:38:10	sumana	24	Female	English	Hyderabad	The product was amazing
1	1/20/2025 0:38:31	Nadeesh	25	Male	English	Vijayawada	The product was too good, loved
2	1/20/2025 0:39:15	Vipul	23	Male	English	Hyderabad	The quality of the product sucks.
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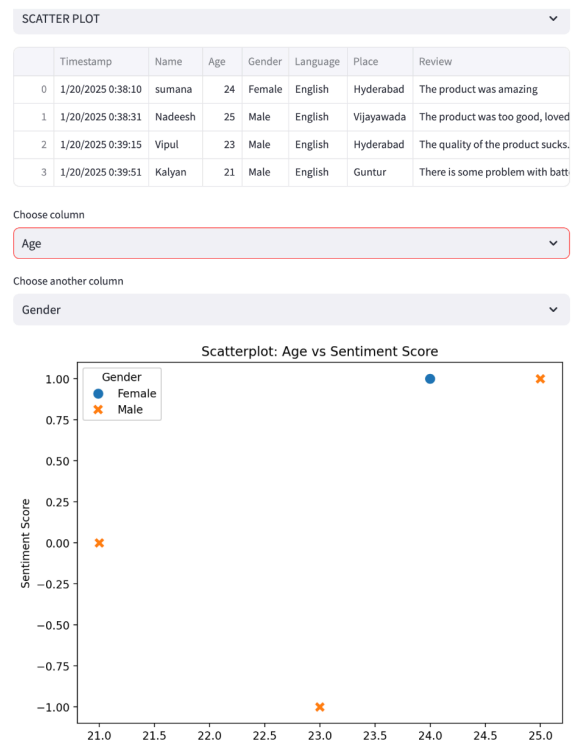
Choose column

Gender



My Menu

RESULTS



Conclusion

The Sentiment Analysis System is a robust application that leverages NLP and data visualization to deliver actionable insights. Its integration with Google Sheets and dynamic visualization capabilities make it an ideal tool for real-time sentiment analysis across various domains, including feedback analysis, survey monitoring, and social media tracking.